Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	**Universidad de Valladolid**
Short name / acronym used in projects	UVa
Participant Identification Code (PIC), if available	999862809.
Website(s)	www.ioba.es
Country	SPAIN
Organisation type	x Academic ☐ Non-academic x Hospital / clinic ☐ SME ☐ Large company x Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Edificio IOBA, Campus Miguel Delibes, Paseo de Belén 17 47011 Valladolid, SPAIN
Department(s), institute(s), research group(s) or unit(s) that would be involved	Biomarkers in ocular surface inflammation and ocular pain Group.
Additional institutional resources relevant to collaboration	- Facilities and equipment from IOBA-Uva: https://www.ioba.es/investigacion/laboratorios-y-unidades-de-investigacion-clinica/ - Centralized services of Uva: https://aplicaciones.educa.jcyl.es/infrared/welcome.action - ICTS Nanbiosis (CIBER-BBN): https://www.nanbiosis.es/

Main contact person for this profile	**María Jesús González García, OD, PhD. University of Valladolid Full Professor**
Main contact email	mariajesus.gonzalez.garcia@uva.es
Other contact person(s), if relevant	Amalia Enríquez-de-Salamanca, PhD. University of Valladolid, Senior Researcher. amalia.enriquez-salamanca@uva.es

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
x Ocular surface	☐ Retina
☐ **Inherited retinal dystrophies (IRDs)**	☐ Age-related ocular disease
☐ **Neuro-vision / integrated vision / neuro-ophthalmology**	☐ Optometry / visual behaviour / rehabilitation
☐ **Advanced therapies / regenerative medicine**	☐ Gene therapy / cell therapy / tissue engineering
x Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ **Drug discovery / screening / repurposing**	☐ Ocular vascular disease
☐ **Infection / trauma-related ocular disease**	x Ocular pain / quality of life / patient-reported outcomes
x Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
x Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database **(please provide up to 10)**	**Ocular Surface Inflammation; Dry Eye; Ocular Pain; Tear Biomarkers; Contact Lens; Conjunctiva; Gene and miRNA Expression;**
Geographies of interest for exchanges **(optional)**	

Overview of your research / innovation focus

Our research line, entitled "Biomarkers in Ocular Surface Inflammation and Ocular Pain," is dedicated to the comprehensive phenotyping of patients presenting ocular surface (OS) inflammatory disorders and related alterations, alongside the identification and characterization of molecular biomarkers. These include diagnostic, activity-related, therapeutic, predictive, prognostic, and susceptibility markers, primarily assessed in tear fluid and conjunctival epithelial cells. The work focuses on immune-mediated, chronic inflammatory conditions of the ocular surface, such as Dry Eye Disease (DED), ocular pain syndromes, and contact lens discomfort (CLD), which, due to their severity and/or chronicity, may compromise visual function. A central objective of this research is the development of targeted therapeutic strategies to mitigate these conditions.

Our approach is aligned with the principles of Personalized/Precision Medicine, aiming to elucidate the inflammatory and molecular mechanisms underpinning ocular surface disorders. This is achieved through integrated analyses of cytokine and chemokine profiles in tears, together with gene and microRNA expression in conjunctival epithelial cells, combined with detailed clinical phenotyping of patients.

Our group has significantly contributed to the characterization of cytokine and chemokine expression profiles in tear samples from both healthy individuals and patients, particularly those with DED. These findings have reinforced the concept of an inflammatory component even in moderate or evaporative forms of the disease. Furthermore, we have developed predictive models based on tear biomarkers and gene expression signatures in conjunctival epithelial cells, with demonstrated applicability in DED and in severe chronic ocular manifestations of graft-versus-host disease.

Additionally, our research has advanced the understanding of molecular mechanisms underlying ocular pain, both in DED patients and in post-surgical corneal conditions. In the field of therapeutics, we have demonstrated the efficacy of several natural polyphenolic compounds—such as resveratrol, quercetin, and hydroxytyrosol—in experimental models, both in vitro and in vivo, supporting their potential as novel treatment options for DED.
Finally, our group is actively engaged in the development of molecular-based point-of-care diagnostic devices, designed to facilitate the translation of biomarker research into clinical practice and improve patient management through rapid, personalized assessment.

Why your team could be relevant to a translational eye-health and vision consortium

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.

Ocular surface clinical studies and translational research capabilities:

- Highly trained clinical research team with extensive expertise in comprehensive ocular surface evaluation, ensuring high reproducibility and standardisation across studies.
- Advanced experience in in vivo confocal microscopy (IVCM) for the analysis of corneal nerves, immune cells and microstructural changes in ocular surface disorders.
- Expertise in non-contact corneal esthesiometry for the assessment of corneal sensitivity and neurosensory function, enabling detailed characterisation of ocular surface pain and neurotrophic conditions.
- Proven capability in tear film evaporimetry, allowing quantitative assessment of tear film stability and environmental impact on dry eye disease and ocular surface disorders.
- Established protocols for biological sample collection and processing, including tear sampling and conjunctival impression cytology, supporting biomarker discovery and translational research approaches.
- Integrated work within the clinical area of IOBA (University of Valladolid), with direct access to well-characterised patient cohorts through specialised units in:
    - Ocular surface disease
    - Contact lens-related conditions
    - Oculofacial chronic pain
- Strong experience in patient-oriented clinical studies, including observational and interventional designs, with a focus on dry eye disease, contact lens discomfort and ocular surface pain.
- Unique access to and operation within a Controlled Environmental Laboratory (CELab), enabling:
    - Precise modulation of environmental conditions (temperature, humidity, airflow, atmospheric pressure)
    - Standardised and reproducible exposure scenarios
    - Investigation of environmental triggers and their impact on ocular surface physiology and symptoms
- High translational potential through the integration of clinical phenotyping, objective measurements and biological sampling, facilitating links between clinical outcomes, neurosensory mechanisms and molecular biomarkers.

Biological sample processing and Analysis.

- Highly trained personnel with extensive experience in handling and processing patient-derived samples, including tear fluid and conjunctival impression cytology specimens, ensuring high reproducibility and quality control.
- Expertise in multiplexed immunoassays (Luminex/xMAP platform) for the simultaneous quantification of cytokines, chemokines, and other inflammatory mediators in tear samples, enabling high-throughput and systems-level biomarker profiling.
- Advanced capabilities in quantitative real-time PCR (RT-PCR) for the analysis of gene expression and microRNA signatures in conjunctival epithelial cells, supporting mechanistic insights and biomarker validation.
- Design and execution of mechanistic studies using human ocular surface cell lines, enabling controlled investigation of inflammatory pathways, drug screening, and validation of therapeutic targets.
- Ability to correlate molecular findings with detailed clinical phenotyping, supporting biomarker qualification and the development of predictive and diagnostic models.
- Implementation of harmonized protocols for sample collection (tear and impression cytology), processing, and analysis, ensuring consistency across studies and facilitating inter-institutional collaboration.
- Established workflows that bridge laboratory analyses with clinical application, including the identification of biomarker panels suitable for development into point-of-care diagnostic platforms.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

- **Ocular Surface Unit and Contact Lens Unit of the IOBA Clinical Area,** where study participants are examined. These units are equipped with all the ophthalmic instruments required for patient evaluation, including:
  slit-lamp biomicroscopes, visual acuity measurement systems, refractive equipment, a corneal topographer, a confocal microscope, and a tear evaporimeter.

- **IOBA Clinical Trials Unit**, which provides support regarding ethical and legal requirements for patient evaluation, as well as data and sample storage.

- **Controlled Environment Chamber (CELab)**, which allows precise control of temperature and humidity in the absence of airflow, ensuring that all participants are assessed under identical environmental conditions. This guarantees standardized data collection and prevents variations attributable solely to fluctuations in environmental parameters.

- **IOBA Molecular Biology Laboratory**, which is equipped with:
  - MAGPIX® multianalyte analysis system and X-Ponent® 3.0 software based on X-MAP technology (Luminex Corporation), as well as the BeadView (Upstate-Millipore) and Belysa (Millipore, Merck) analysis software for multiplex cytokine assays.
  - SpectraMax spectrophotometer and SoftMax® software for ELISA-based determinations, as well as all laboratory materials required for processing and measuring tear samples.
  - Chemidoc MP (BioRad) equipment and ImageLab (BioRad) software for Western Blot protein analysis
  - QuantiStudio 5 (Applied Biosystems) real time-PCR thermoclycler, for gene and miRNA expression analysis.
  - Fluorescence microscope Leica (Leica DMI 6000 B, Leica).

- **IOBA Cell culture Laboratory**, equipped with all necessary equipment for cell culture, including vertical flow cabins and incubators.

- **Instrumental Techniques Laboratory (LTI-INFRARED)**: for confocal microscopy and flow cytometry studies (LTI-INFRARED Platform, https://aplicaciones.educa.jcyl.es/infrared/welcome.action).

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

- **Structured training environment** for early-stage and postdoctoral researchers, embedded within a multidisciplinary clinical and research setting at IOBA (University of Valladolid), ensuring exposure to both basic science and patient-oriented research.
- **Training in translational ocular surface research**, enabling researchers to integrate clinical phenotyping, advanced diagnostic techniques and molecular biomarker analysis within a precision medicine framework.
- **Hands-on training in advanced clinical and diagnostic techniques**, including:
  - In vivo confocal microscopy (IVCM) for corneal nerve and immune cell analysis
  - Non-contact corneal esthesiometry for neurosensory function assessment
  - Tear film evaporimetry and ocular surface functional testing
- **Advanced training in biological sample collection and processing**, including tear sampling and conjunctival impression cytology, linked to downstream molecular analyses (protein, gene and miRNA expression), including:
  - Hands-on expertise in protein extraction, quantification, and characterization using techniques such as Western blotting, spectrophotometric methods, immunofluorescence, ELISA, and multiplexed Luminex xMAP® technology for high-throughput, multi-analyte profiling.
  - Gene and miRNA expression analysis: Comprehensive training in nucleic acid workflows, including mRNA and miRNA extraction, quantification, and quantitative real-time RT-PCR, enabling precise characterization of molecular signatures associated with ocular surface diseases.
- **Exposure to controlled-environment experimental research** through the CELab (Controlled Environmental Laboratory), allowing incoming staff to design and perform studies under strictly regulated environmental conditions.
- **Strong capacity to train researchers in patient-based clinical studies**, including study design, protocol implementation, ethical considerations and interaction with well-characterised cohorts from specialised units (ocular surface disease, contact lens, and oculofacial chronic pain).
- **Training in translational research approaches**, bridging clinical data with molecular biomarkers and supporting the development of diagnostic tools and therapeutic strategies.

- **Opportunities for interdisciplinary knowledge exchange**, including interaction with clinicians, biologists, optometrists and data analysts.
- **Ocular surface cell culture models**: Training in the maintenance and experimental use of human ocular surface cell lines, including the design of in vitro assays to investigate inflammatory mechanisms and evaluate therapeutic compounds.

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. Again, please think broadly and beyond ophthalmology.

- Access to complementary cutting-edge methodologies and technologies not currently available within our institution, particularly in areas such as proteomics, advanced multi-omics integration, AI-assisted data analysis and digital health tools applied to ocular surface disease.
- Strengthening of translational research pathways, enabling a more efficient progression from biomarker discovery to clinical validation and eventual implementation in diagnostic and therapeutic strategies.
- Opportunities to collaborate with non-academic partners (SMEs, industry, and technology developers), particularly in the areas of point-of-care diagnostic devices, biomarker-based tools and innovation-oriented product development.
- Expansion of our expertise in patient-centred research approaches, including the development and validation of Patient-Reported Outcome Measures (PROMs) and their integration with objective clinical and molecular data.
- Access to diverse patient populations and clinical settings, enhancing the robustness and external validation of our research findings.
- Enhancement of our capabilities in data science, bioinformatics and advanced statistical modelling, particularly for the integration of multidimensional datasets (clinical, molecular and environmental).
- Exposure to different regulatory, ethical and clinical frameworks across countries, improving our capacity to design and implement multicentre international clinical studies.
- Strengthening long-term strategic collaborations and international networks, facilitating joint publications, future EU proposals and sustainable knowledge exchange beyond the duration of the project.
- Training and career development opportunities for our researchers through intersectoral and interdisciplinary mobility, particularly benefiting early-career and postdoctoral researchers.
- Increased capacity to co-develop and validate environment-sensitive research paradigms, building on our Controlled Environmental Laboratory (CELab) expertise through collaboration with partners working on environmental health, exposure science and related fields.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Maria Jesús González García	Full Professor. Principal Investigator	IOBA-UVa	Researcher	Experienced	mariajesus.gonzalez.garcia@uva.es
Amalia Enríquez de Salamanca Aladro	Senior Scientist. Principal Investigator	IOBA-UVa	Researcher	Experienced	amalia.enriquez-salamanca@uva.es
Alberto López Miguel	Full Professor, Principal Investigator	IOBA-UVa	Researcher	Experienced	alberto.lopez.miguel@uva.es
Itziar Fernández Martínez	Associate Professor / Principal Investigator	IOBA-UVa	Researcher	Experienced	itziar.fernandez@uva.es
Marta Blanco Vázquez		IOBA-UVa	Researcher	Middle-Stage	marta.blanco.vazquez@uva.es
Alberto López de la Rosa	Associate Professor	IOBA-UVa	Researcher	Experienced	alberto.lopez.rosa@uva.es
Andrés Ángel Calderón García	Assistant Professor	IOBA-UVa	Researcher	Middle-Stage	andresangel.calderon.garcia@uva.es

Elena Martínez Plaza	Postdoctoral researcher	IOBA-UVa	Researcher	Middle-Stage	elena.martinez.plaza@uva.es
Cristina Arroyo del Arroyo	Postdoctoral researcher	IOBA-UVa	Researcher	Middle-Stage	cristina.arroyo.arroyo@uva.es
Cristina Valencia Sandónis	Predoctoral Student	IOBA-UVa	Researcher	Early Stage	cristina.valencia@uva.es
Ivonne Guachamin	Predoctoral Student	IOBA-UVa	Researcher	Early Stage	ivonne.guachamin25@estudiantes.uva.es
Lidia Cocho Archiles	Ophtalmologist	HCU/ IOBA-UVa	Opthalmologists	Experienced	lidiacocho@hotmail.com

Main disciplines represented in your team	**e.g. ophthalmology, optometry, cell biology, immunology, genetics, biomaterials, data science, engineering, AI, regulatory affairs, marketing and commercialisation.** Ophthalmology, Optometry, Biochemistry, Molecular Biology, Cell biology, Biostatistics/Data science, Clinical trials,
Experience in EU / international / multi-partner collaborative projects	• Implementation of molecular biomarker-based biosensor for patient stratification and tailored therapy in inflammatory-dry eye disease. Stratif-EYE. Proposal number 754620-1. Entidad financiadora: Programa H2020. Call H2020-SC1-2017. (not granted). Coordinators: CIBER BBN IOBA-UVa CB06/01/0003 Group: Leader: A. Enríquez-de-Salamanca (IOBA). Partners: 1) Centro De investigación biomédica En Red (España) (Coordinadores); 2) Tampereen Yliopisto (Finlandia); 3) Klinikum Der Universitaet Zu Koeln (Alemania); 4) Vision I Mas D Sl (España); 5) Asper Biotech As (Estonia); 6) Asphalion Sl (España); 7) Innova Srl (Italia); 8) Asociación Española Síndrome De Sjögren (Aess) (España); 9) Bundesverband-Auge E.V (Alemania • IT-DED 3 MSC EU project. María Jesus González-García UVa Pirncipal investigator; WP4 Leader,ERS6 Co-Supervisor. Amalia Enríquez-de-Salamanca WP1 Leader; ESR4 Co supervisor; Supervisory board member. • CHOPIN MSC Doctoral Network EU project 2024, 2025 and 2026 calls application: Not Granted • COST ACTION 2025. Granted (ACTION CA 24112 "Tear precision", 2025-2029).(www.tear-precision.eu) • Tear Research Network (TRN; www. Tearresearchnetwork.com)
Experience with clinical, translational or patient-facing research (if relevant)	x Yes ☐ No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

- Vázquez A, Martínez-Plaza E, Fernández I, Sobas EM, González-García MJ, Enríquez-de-Salamanca A, Ortega E, López-Miguel A, Calonge M. Phenotypic characterization of patients developing chronic dry eye and pain after refractive surgery: A cross-sectional study. Ocul Surf. 2022 Oct;26:63-74. doi: 10.1016/j.jtos.2022.07.010.

- Calderón-García AÁ, Valencia-Nieto L, Valencia-Sandonis C, López-de la Rosa A, Blanco-Vazquez M, Fernández I, García-Vázquez C, Arroyo-Del Arroyo C, González-García MJ, Enríquez-de-Salamanca A . Gene expression changes in conjunctival cells associated with contact lens wear and discomfort. Ocul Surf. 2024 Jan;31:31-42. doi: 10.1016/j.jtos.2023.12.004

- Calderón-García AÁ, López-de la Rosa A, Valencia-Nieto L, García-Vázquez C, Arroyo-Del-Arroyo C, López-Miguel A, Blanco-Vázquez M, Novo-Diez A, Fernández I, Enríquez de Salamanca A, González-García MJ . Tear Cytokines as Predictive Biomarkers of Success in Contact Lens Discomfort Management. Ocul Immunol Inflamm. 2024 Dec;32(10):2339-2347. doi: 10.1080/09273948.2024.2361114

- Blanco-Vázquez, M.;Novo-Diez, A.; Vázquez, A.; Enríquez-de-Salamanca, A.;González-García, M.J.; Calonge, M. Clinical Characterization of the Lacrimal Functional Unit in Patients with Chronic Ocular Pain Associated with Dry Eye Disease. J. Clin. Med.2025, 14, 5250. https://doi.org/10.3390/jcm14155250.

- Blanco-Vázquez M, Vázquez A, Fernández I, Novo-Diez A, Martínez-Plaza E, García-Vázquez C, González-García MJ, Sobas EM, Calonge M, Enríquez-de-Salamanca A . Inflammation-related molecules in tears of patients with chronic ocular pain and dry eye disease. Exp Eye Res. 2022 Jun;219:109057. doi: 10.1016/j.exer.2022.109057.

- Cocho L, Fernández I, Calonge M, Sainz de la Maza M, Rovira M, Stern ME, Garcia-Vazquez C, Enríquez-de-Salamanca A. Prehematopoietic Stem Cell Transplantation Tear Cytokines as Potential Susceptibility Biomarkers for Ocular Chronic Graft-Versus-Host Disease. Invest Ophthalmol Vis Sci. 2017 Sep 1;58(11):4836-4846. doi: 10.1167/iovs.17-21670.

Up to 5 relevant previous projects, grants, networks or collaborations

1) **IT-DED 3 MSC EU project (INTERNATIONAL)**
Integrated Training in Dry Eye Disease Drug Development (IT-DED3)
Grant agreement number 765608
Entidad financiadora: EUROPA PROGRAMA H2020
Call: H2020-MSCA-ITN-2017; Funding scheme: MSCA-ITN-ETN
2018-2022
- Coordinators: UNIVERSITEIT ANTWERPEN (BELGICA).
- Beneficiaries: UNIVERSITEIT ANTWERPEN (BELGICA); Universite Pierre Et Marie Curie - Paris 6 (Francia); UNIVERSIDAD DE VALLADOLID (ESPAÑA) : Maria J González García: UVA IP,WP4 Training leader; ESR3 co-supervisor. Amalia Enríquez de Salamanca: WP1 Drug Discovery leader; Supervisory board member; ESR4 Co-supervisor

Latvijas Organiskas Sintezes Instituts (Letonia); Ita-Suomen Yliopisto (Finlandia); Klinikum Der Universitaet Zu Koeln (Alemania); Instituto De Biologia Experimental E Tecnologica (Portugal)

- Partners: Quintessence Consulting NV; Mercachem BV; HG Beyond; Riga Technical University; New University of Lisbon; Novaliq GmbH DE; Centre Hospitalier National des Quinze-Vingts; Horus Pharma; Heildelberg Enginieering.

Total amount granted: 3,002,689.44€ (UVA : 495,745.92€)

2) **COST Action CA24211 (INTERNATIONAL):**
Implementation of Tear Fluid Biomarkers in Precision Medicine. TEAR-Precision (COST ACTION; Open Call Collection OC-2024-1
Proposal Reference OC-2024-1-27225
REF : Action N. CA24112
COST European Cooperation in Science and technology. COST Association AISBL. European Union.
2025-2029.4 años
IP: Marlies Gijs. Maastricht University.
Main Proposer: The Netherlands
Network of Proposers:
-Full Member: Belgium, Croatia, Czech Republic, Denmark, Estonia, France, Germany, Greece, Hungary,Italy, Lithuania, Moldova, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Spain, Sweden,Switzerland, Türkiye, United Kingdom.
- Cooperating Member: Israel
- International Partner: Australia, Canada, Colombia, Hong Kong SAR, India, New Zealand, Singapore,United States
- Specific Organisations :EU Institutions, Bodies, Offices and Agencies (EC/EU): University of Crete Medical School
- International Organisation: Association for Research in Vision and Ophthalmology; Schepens Eye Research Institute; Schepens Eye Research Institute; Tan Tock Seng Hospital; Vision Research Foundation, Sankara Nethralaya; Cincinnati Children's Hospital Medical Center
-Industrial Dimension: SMEs: Canada, Netherlands, Switzerland, United States. Large companies: Switzerland
Spain: Amalia Enríquez de Salamanca (FUNGE), MC member for Spain; (WG 4 co-lider); María J González-García (UVA)

Amount granted: 600000€

3) **CIBER-BBN (Bioengineering, Biomaterials and Nanomedicine Research Network)(National)**
IOBA-UVa CB06/01/0003 Group. Instituto de Salud Carlos III, Ministerio de Sanidad y Consumo.
Group Leader Margarita Calonge Cano.
01/2007-actualidad
https://www.ciber-bbn.es
The CIBER in its thematic area of Bioengineering, Biomaterials and Nanomedicine (CIBER-BBN) is currently composed of 45 research groups (43 full members, 1 associated group, and 1 linked clinical group), selected on the basis of their scientific excellence. These groups work primarily within three scientific programs: 1) Bioengineering and Medical Imaging, 2) Biomaterials and Advanced Therapies, and 3) Nanomedicine. The center's activities are aimed both at developing systems for prevention, diagnosis, and monitoring, and at advancing technologies related to specific therapies, such as regenerative medicine and nanotherapies. CIBER-BBN has been carrying out its work since 2007, providing support, training, infrastructures, and resources to researchers, thereby contributing to the promotion of scientific research in our country. The overarching goals outlined in the CIBER-BBN Strategic Plan for the 2018–2021 period include maintaining the scientific and technological excellence of the thematic area, fostering collaborative research among CIBER-BBN groups, promoting translational research, and facilitating the transfer of research results to the productive sector.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Please indicate only what can be shared at this stage.

1. **Title: COMPOSICIÓN PARA SU USO EN EL TRATAMIENTO Y/O PREVENCIÓN DE LA INFLAMACIÓN, EL ESTRÉS OXIDATIVO Y LA NEOVASCULARIZACIÓN OCULAR**
   REF: ES 2 428 665 B1
   Inventores: González García, MJ; Calonge Cano M, Enríquez-de-Salamanca Aladro, A; Stern, ME; Avengózar Vela, A.
   N. de solicitud: P201230664 País de prioridad: España Fecha de prioridad: 04-Mayo 2012
   Entidad titular: Universidad de Valladolid, Valladolid España; Allergan Inc,California, EEUU

2. **Registro de la propiedad intelectual del texto científico "Contact Lens Discomfort Index" ,** Universidad de Valladolid.
   Autores (p.o. de firma): Arroyo del Arroyo, C; Fernández Martínez, I; López-Miguel A; González García, MJ, López de la Rosa, A; Enríquez-de-Salamanca Aladro, A
   01/03/2019
   N. de solicitud: VA-71-2019 País de prioridad: España

3. **Title: COMPUESTOS PARA SU USO EN LA PREVENCIÓN Y/O TRATAMIENTO DE LA INFLAMACIÓN Y EL ESTRÉS OXIDATIVO DEL SEGMENTO ANTERIOR DEL OJO.**
   Inventores (p.o. de firma): Margarita Calonge Cano, Maria José Cocero, María Jesús González García, SORAYA Rodriguez Rojo , Nikolaos Katsinas, Enríquez-de-Salamanca Aladro, A;;.
   REF : ES2080.29
   N. de solicitud P202031306 País de prioridad: España Fecha de depósito: 28 diciembre 2020
   Entidad titular: Universidad de Valladolid, Valladolid España.

4. **Clinical Protocol: "Enfermedad de ojo seco: detección y manejo",** González García, MJ, Blanco Vázquez M, Arroyo del Arroyo C, Calonge Cano M.. Editorial: Grupo ICM de comunicación, Madrid 2026. ISBN: 979-13-990617-8-9.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	**x Yes ☐ Maybe ☐ No capacity to send staff on secondment**
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers x Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles

Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	e.g. visa, insurance, access to facilities, language, employment status, health & safety induction

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice

Please summarise the 2-3 strengths that should be first visible in an internal searchable profile.

- **Strong translational and multidisciplinary capacity in ocular surface diseases**: Integration of clinical expertise (ophthalmology and optometry) with molecular biology, biochemistry, and biostatistics, with a specific focus on Dry Eye Disease, contact lens discomfort, and ocular pain, enabling robust patient phenotyping and direct linkage to underlying molecular mechanisms.
- **Advanced expertise in biomarker discovery and molecular analysis**: Proven capabilities in tear and conjunctival sample processing and analysis using multiplex xMAP (Luminex), ELISA, and RT-PCR for gene and miRNA profiling, supporting the identification and validation of clinically relevant biomarkers in dry eye, contact lens-related conditions, and ocular pain.
- **Established experimental and innovation platform with translational impact**: Extensive experience in in vitro ocular surface models and therapeutic evaluation, particularly in Dry Eye Disease, combined with the development of molecular-based point-of-care diagnostic tools to facilitate clinical implementation and personalized management.

Capabilities or expertise you would especially like to access through the consortium

Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium.

We would like to connect with other research teams or clinical units with expertise in or related to;

- Proteomic analysis facilities.
- Pain units
- Ocular Surface research/clinical groups.
- Point of Care development companies
- Patient-related output measurements (PROMs) development
- Data mining/analysis

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	**x Yes** ☐ **No**

x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.

Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

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x Understood

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